

How Do Stock Markets React to Natural Disasters?

A Comparative Analysis of the 2011 Great East Japan Earthquake, 2005 Hurricane Katrina, and the 2023 Türkiye Earthquakes

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Abstract

This paper examines how equity markets respond to major natural disasters by comparing three events: the 2011 Great East Japan Earthquake, Hurricane Katrina in 2005, and the 2023 Türkiye earthquakes. Using benchmark stock indices (the Nikkei 225, S&P 500, and BIST 100) together with data on disaster severity, institutional responses, and sectoral exposure, the study investigates whether a disaster's human and economic toll predicts the severity of its impact on financial markets.

Drawing on eight academic studies alongside official reports from the World Bank, IMF, central banks, and other primary sources, the analysis finds that stock market reactions do not closely correlate with death tolls or the number of people affected. Instead, the depth and duration of market declines are more strongly influenced by a country's economic structure, the concentration of damage in key industries, and the speed and credibility of monetary, fiscal, and regulatory responses. The Türkiye case also demonstrates that regulatory intervention can directly influence market outcomes by interrupting normal price discovery.

Overall, the findings suggest that financial market resilience depends less on the magnitude of a disaster itself and more on the strength and structure of the economy it affects.

Introduction

Natural disasters destroy physical infrastructure, displace populations, and place enormous pressure on government finances. However, stock markets are forward-looking pricing mechanisms rather than simple measures of human loss. A country may experience a devastating humanitarian disaster while its stock market records only a modest decline, whereas a less severe disaster elsewhere may produce a much larger and longer-lasting market reaction. This difference between a disaster's human impact and its financial impact is the central question investigated in this paper.

The 2011 Great East Japan Earthquake, Hurricane Katrina in 2005, and the 2023 Türkiye earthquakes provide a useful basis for comparison because they affected economies with very different structures. Japan was a high-income, manufacturing-based economy facing long-term economic stagnation. The United States was the world's largest and most diversified economy, while Türkiye was a middle-income economy already experiencing high inflation and currency instability. Each disaster also prompted a different institutional response—from aggressive monetary easing in Japan, to continued monetary tightening in the United States, and the temporary closure of the Turkish stock exchange during the market sell-off.

This paper argues that these structural and institutional differences, rather than the size of the disasters themselves, explain the different market outcomes. To examine this argument, the study compares the performance of each country's benchmark stock index—the Nikkei 225, S&P 500, and BIST 100—using a common set of variables: economic damage as a

share of GDP, fiscal and monetary responses, insurance systems, sectoral exposure, and pre-disaster macroeconomic conditions.

The paper is organised into six main sections. It begins with a review of the existing literature on financial market reactions to natural disasters before presenting the research question, hypothesis, and methodology. Three detailed case studies then examine each disaster individually. These are followed by a comparative analysis, a discussion of the key findings, the study's limitations, and a final conclusion.

Literature Review

The relationship between natural disasters and stock market performance has been widely studied in finance, particularly through the Efficient Market Hypothesis (EMH). According to Fama (1970), financial markets quickly incorporate publicly available information into asset prices. From this perspective, a natural disaster should affect stock prices only to the extent that it changes investors' expectations about future economic performance, rather than because of the disaster's human impact alone. To measure these effects, researchers commonly use the event study method developed by Fama, Fisher, Jensen, and Roll (1969), which examines abnormal stock returns around a significant event. This methodology forms the basis of most research on disasters and financial markets and provides the analytical framework for this paper.

Research on the 2011 Great East Japan Earthquake suggests that disaster impacts vary considerably across industries. Luo (2012) found that although the earthquake generated negative shocks in several global stock markets, the overall impact on broad market indices outside Japan was relatively small and statistically insignificant. Instead, the strongest effects were concentrated in industries directly linked to the disaster. Similarly, Wang and Kutan (2013) used GARCH models to examine Japanese and U.S. markets and found little evidence of a significant aggregate wealth effect. However, they observed substantial differences within the insurance sector, where Japanese insurers experienced positive abnormal returns while U.S. insurers recorded losses. These studies suggest that broad market indices often recover quickly, even though individual sectors may experience significant disruptions.

Research on Hurricane Katrina reaches similar conclusions. Howerton and Bacon (2017) found that the hurricane significantly affected property and casualty insurance companies, while the wider U.S. stock market remained relatively stable. They argued that investors had already incorporated many of the storm's expected costs into prices before landfall, providing evidence in support of the semi-strong form of market efficiency. Feria-Domínguez, Paneque, and Gil-Hurtado (2017) extended this analysis by examining seven major Atlantic hurricanes between 2005 and 2012. Their findings showed that most hurricanes generated significant abnormal returns within the property and casualty insurance sector. Interestingly, Katrina itself produced weaker abnormal returns than expected, which the authors attributed to the market's anticipation of the disaster before it occurred.

More recent research on the 2023 Türkiye earthquakes adds another important perspective by examining firm-level responses. Pandey, Kumar, Kumari, and Alahdal (2024) analysed 382 firms listed on Borsa Istanbul and found significantly negative abnormal returns during

the first three trading days after the earthquake. However, returns became positive on the fourth and fifth trading days, with firms in the Basic Materials sector performing particularly well due to expectations of increased reconstruction activity. This demonstrates that even within a single economy, different industries can respond very differently to the same disaster.

A broader perspective is provided by Bourdeau-Brien and Kryzanowski (2017), who analysed numerous natural disasters across the United States. Their research found that most disasters had only limited effects on overall market returns but produced significant negative impacts on firms located within the affected regions. They also reported that return volatility for these firms more than doubled during the months following a disaster, highlighting the importance of geographical exposure alongside industry characteristics.

Taken together, the existing literature points to two consistent conclusions. First, broad stock market indices are generally more resilient to natural disasters than many investors initially expect. Second, the greatest financial impacts tend to occur within specific industries and firms rather than across the entire market. However, most previous studies focus on individual disasters or single countries. Comparatively little research examines how differences in economic structure, institutional strength, and policy responses influence market reactions across countries. This paper seeks to address that gap by comparing three major disasters that occurred under very different economic and institutional conditions.

Methodology

This study uses a comparative case study approach to examine how stock markets responded to three major natural disasters: the 2011 Great East Japan Earthquake, Hurricane Katrina in 2005, and the 2023 Türkiye earthquakes. A comparative approach was chosen because it allows similarities and differences between the three cases to be analysed in detail. While examining only three disasters does not allow for broad statistical conclusions, it provides an opportunity to explore how economic structure, institutional responses, and sectoral exposure may influence market behaviour following catastrophic events.

The three case studies were selected because they represent economies with very different characteristics. Japan was a high-income economy with a strong manufacturing base and a high debt-to-GDP ratio. The United States was the world's largest and most diversified economy, while Türkiye was a middle-income economy experiencing high inflation and currency instability before the earthquakes. These differences make the three disasters suitable for comparison.

For each case, benchmark stock market data were collected, including the closing price on the last trading day before the disaster, the first closing price after the event, the lowest point reached during the market decline, and the date on which the index returned to its pre-disaster level. The benchmark indices used were the Nikkei 225 for Japan, the S&P 500 for the United States, and the BIST 100 for Türkiye.

To understand why market reactions differed, these financial indicators were analysed alongside several economic and institutional variables. These included the number of deaths, the number of people affected, direct economic damage as a percentage of GDP, fiscal stimulus measures, central bank responses, catastrophe insurance systems, sectoral exposure, and pre-disaster macroeconomic conditions. Data were collected from official sources, including the World Bank, the International Monetary Fund, national governments, central banks, and recognised financial institutions.

The study also examines how the structure of each economy influenced market performance. Particular attention is given to whether the industries most heavily affected by each disaster represented nationally important sectors with significant weights in the broader economy. This helps explain why disasters with similar economic costs or human impacts produced different stock market outcomes.

The literature review draws on six peer-reviewed studies that examine stock market reactions to natural disasters, supported by two foundational papers on market efficiency and event study methodology. Together, these sources provide the theoretical framework for analysing the three case studies and interpreting the findings.

This research is primarily qualitative rather than statistical. Instead of testing the hypothesis using large datasets or econometric models, the study compares evidence from the three case studies to determine whether the observed patterns are more consistent with the proposed hypothesis (H1) or the alternative hypothesis (H0). Although this approach cannot establish causation with certainty, it provides a structured and evidence-based comparison of how different economies respond to major natural disasters.

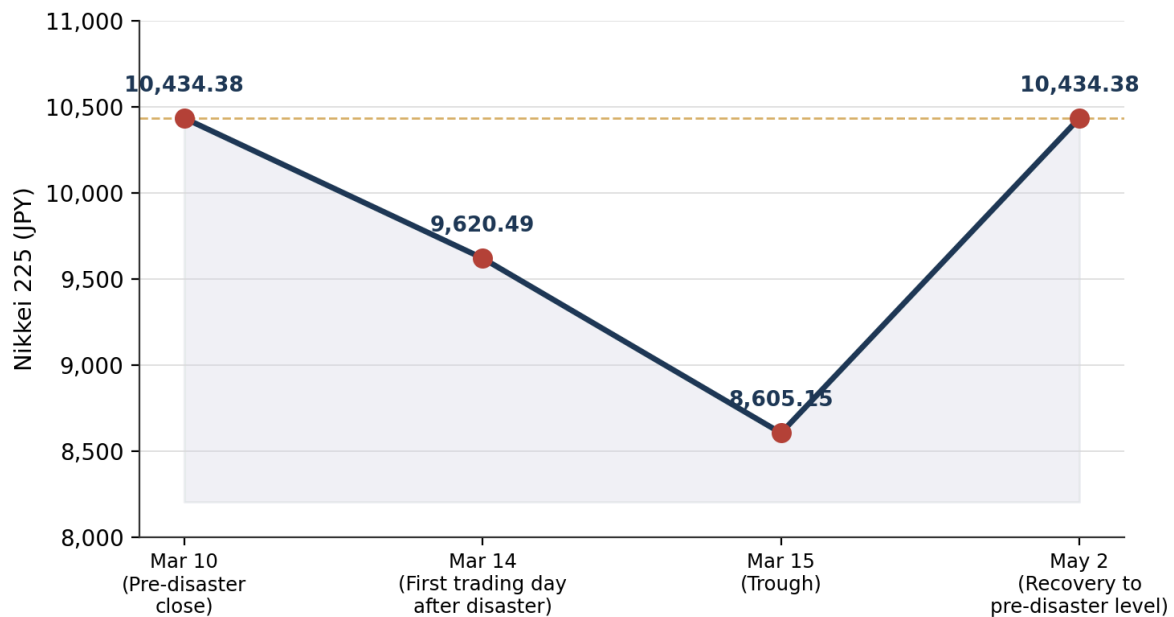
Case Study 1: The 2011 Great East Japan Earthquake

The Great East Japan Earthquake struck on 11 March 2011 and triggered a devastating tsunami that led to the Fukushima Daiichi nuclear accident. Rather than being a single disaster, it became a cascading crisis involving an earthquake, a tsunami, and a nuclear emergency. The Nikkei 225 closed at **10,434.38 JPY** on 10 March 2011, the final trading day before the disaster.

When the market reopened on 14 March, the Nikkei 225 had fallen to **9,620.49 JPY**. Selling continued over the following day, with the index reaching a low of **8,605.15 JPY** on 15 March. This represented a peak-to-trough decline of approximately **17.5%**, the largest decline among the three case studies. The index eventually recovered to its pre-disaster

level on **2 May 2011**, around seven weeks after the earthquake.

Figure 1. Nikkei 225 Reaction to the 2011 Great East Japan Earthquake



Note. Stylized trajectory connecting the four key reported data points; not a full daily series.

The Japanese government's emergency response began immediately after the disaster. A nuclear emergency was declared within hours, evacuation orders were issued around the Fukushima Daiichi plant, and the Self-Defense Forces, police, and emergency services were rapidly deployed to the affected regions. Within days, Sendai Airport had reopened for emergency relief operations, allowing international aid and supplies to reach the disaster zone more efficiently. Two weeks after the earthquake, twenty international search-and-rescue teams from fifteen countries were operating across the affected region.

Table 1

Japan Emergency Response Timeline

Time After Disaster	Government and Institutional Action
0–5 hours	Article 15 nuclear emergency declared; evacuation orders issued around Fukushima
First 24 hours	Self-Defense Forces, police, and firefighters deployed across Tōhoku

4 days	Sendai Airport reopened for emergency relief operations
2 weeks	Twenty international search-and-rescue teams from fifteen countries deployed

Financial authorities also acted quickly to stabilise markets. On the day of the disaster, the Bank of Japan injected **¥15.1 trillion** into the financial system to maintain liquidity. It later expanded its asset purchase programme by **¥5 trillion**, followed by another **¥5 trillion**, increasing purchases of commercial paper, corporate bonds, exchange-traded funds (ETFs), and real estate investment trusts (REITs). On 18 March, the Bank of Japan also coordinated with the G7 to stabilise the rapidly appreciating yen. At the same time, the Japanese government approved reconstruction spending that eventually reached approximately **¥25 trillion**.

Despite this rapid and substantial response, Japan experienced the deepest and longest-lasting market decline among the three countries examined. The explanation appears to lie less in the effectiveness of the government's response and more in the structure of the Japanese economy. Manufacturing, which accounted for approximately **18-20% of GDP**, suffered major disruptions as damaged factories interrupted global supply chains. Toyota temporarily halted production in both Japan and overseas because of shortages of critical components. East Japan Railway Company also faced extensive infrastructure damage and significant rebuilding costs. The Fukushima nuclear accident further intensified the economic impact by removing almost **30% of Japan's electricity generation**, creating an energy shortage that affected businesses well beyond the disaster zone.

Estimates of the disaster's economic cost vary depending on the source. Japan's Cabinet Office estimated direct losses at approximately **\$210 billion**, while the World Bank placed the figure closer to **\$235 billion**, equivalent to around **4% of Japan's GDP** (World Bank, 2012). Although early forecasts predicted a much larger contraction, the International Monetary Fund later reported that Japan's economy contracted by only **0.7% in 2011**, demonstrating that the long-term economic impact was less severe than many investors had initially expected (Danninger, 2011).

Japan's earthquake insurance system also influenced how financial losses were distributed. Earthquake insurance operated as a public-private partnership designed primarily to protect the financial stability of insurers rather than fully compensate property owners. Coverage was generally limited to **50% of a property's value**, while standard fire insurance policies excluded earthquake and tsunami damage unless additional cover had been purchased. As a result, insurance compensated for **less than 20%** of the total economic losses, leaving households and the government responsible for the majority of reconstruction costs.

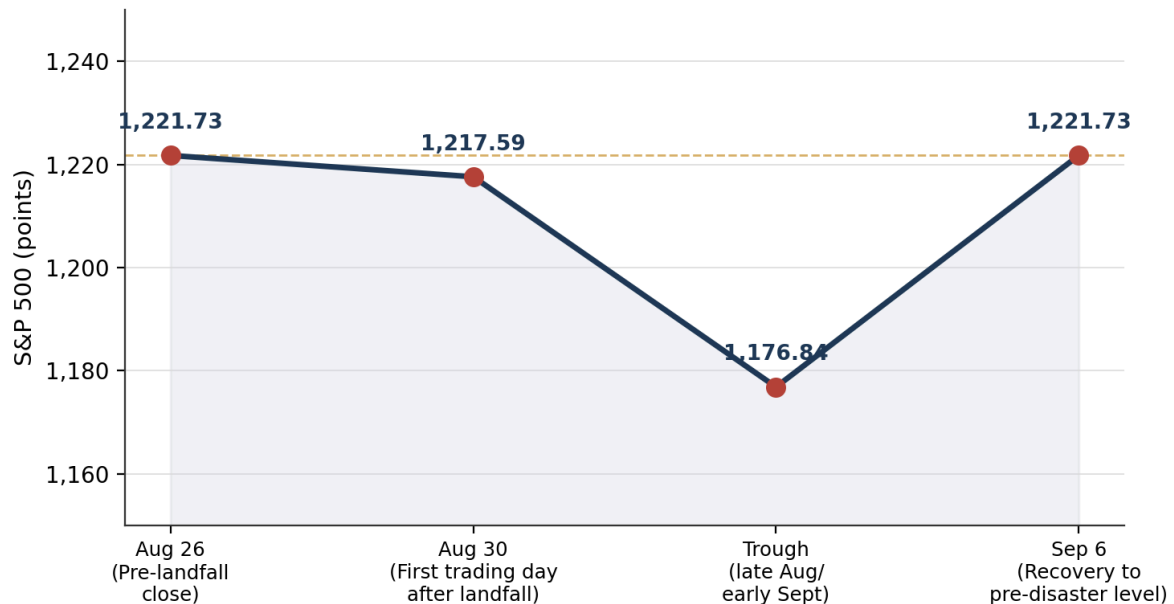
Overall, Japan demonstrates that a rapid and well-funded institutional response does not necessarily prevent a severe market decline. The country's heavy reliance on globally significant manufacturing industries, together with the unprecedented nuclear crisis, created prolonged uncertainty that outweighed the benefits of swift government intervention.

Case Study 2: Hurricane Katrina (2005)

Hurricane Katrina made landfall near New Orleans on **29 August 2005**, becoming one of the costliest natural disasters in United States history. The storm caused catastrophic flooding, widespread infrastructure damage, and significant loss of life along the Gulf Coast. Despite the scale of the humanitarian crisis, its impact on the broader U.S. stock market was comparatively limited. The S&P 500 closed at **1,221.73** on **26 August 2005**, the last trading day before landfall.

When markets reopened after the hurricane, the S&P 500 fell slightly to **1,217.59** on **30 August** before continuing to a low of approximately **1,176.84**. This represented a peak-to-trough decline of around **3.7%**, the smallest decline among the three disasters examined in this study. The index recovered to its pre-disaster level by **6 September 2005**, taking less than two weeks to fully recover.

Figure 2. S&P 500 Reaction to Hurricane Katrina (2005)



Note. Stylized trajectory connecting the four key reported data points; not a full daily series.

The federal government's emergency response received widespread criticism for being slow and poorly coordinated. Flooding caused by levee failures overwhelmed New Orleans, communication systems collapsed, and emergency assistance was delayed despite the growing humanitarian crisis. Large-scale relief operations gradually improved over the following weeks, but the initial response highlighted significant weaknesses in disaster preparedness and coordination.

Table 2

Hurricane Katrina Emergency Response Timeline

Time After Landfall	Government and Institutional Action
First 5 hours	Levee failures cause extensive flooding across New Orleans
First 24 hours	Communications collapse; emergency response hampered by lack of situational awareness
4 days	Major federal supply convoys begin entering New Orleans
2 weeks	Large-scale evacuation completed; FEMA Director Michael Brown resigns

Unlike Japan and Türkiye, the Federal Reserve did not introduce emergency monetary stimulus following the disaster. Instead, it continued its existing policy of gradually raising interest rates to control inflation. At its meeting on **20 September 2005**, the Federal Open Market Committee increased the federal funds rate by **25 basis points** to **3.75%**. Although the Federal Reserve acknowledged that Hurricane Katrina would temporarily reduce spending, production, and employment in the Gulf Coast region, it concluded that the disaster was unlikely to pose a lasting threat to the overall U.S. economy.

The relatively mild market reaction can largely be explained by the size and diversification of the U.S. economy. Katrina caused approximately **\$125 billion** in direct economic damage, but this represented only about **1% of U.S. GDP** at the time. The disaster was also concentrated within the Gulf Coast, allowing most regions and industries across the country to continue operating with limited disruption.

Nevertheless, certain industries experienced significant short-term effects. Oil production in the Gulf of Mexico was heavily disrupted, with around **95%** of production temporarily shut down. The Port of New Orleans, which handled a substantial share of U.S. grain exports, also ceased normal operations. While energy prices increased sharply, many large energy companies recovered quickly as higher fuel prices boosted revenues. Homebuilding companies benefited from increased demand during the reconstruction phase, while property and casualty insurers experienced the largest financial losses. Howerton and Bacon

(2017) found that insurance companies recorded significant abnormal returns even though the broader market remained relatively stable.

The disaster also exposed weaknesses in the U.S. catastrophe insurance system. Wind damage was generally covered by private insurers, whereas flood damage fell under the federally managed National Flood Insurance Program (NFIP). This division created numerous disputes over insurance claims and contributed to the NFIP accumulating a deficit of approximately **\$17 billion**, requiring substantial federal financial support.

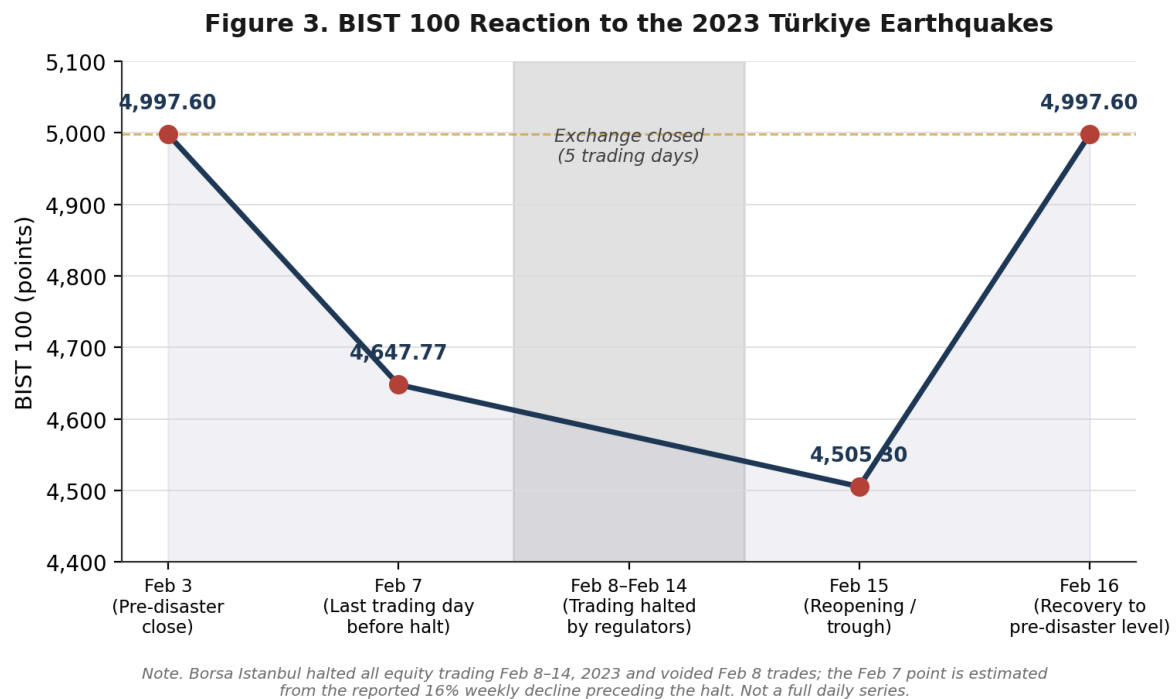
Overall, Hurricane Katrina demonstrates that a disaster's human and economic costs do not necessarily determine the scale of a stock market reaction. Despite a slower institutional response than either Japan or Türkiye, the U.S. market recovered quickly because the damage remained geographically concentrated within a large and highly diversified economy. This supports the paper's central argument that economic structure plays a more important role in determining market resilience than the severity of the disaster itself.

Case Study 3: The 2023 Türkiye Earthquakes

The Türkiye earthquakes of **6 February 2023** were among the deadliest natural disasters of the twenty-first century. The twin earthquakes caused widespread destruction across southern Türkiye, killing **over 50,000 people** and affecting more than **16 million**. In terms of human impact, they were by far the most severe of the three disasters examined in this study. However, the reaction of the Turkish stock market was neither the largest nor the longest-lasting.

The BIST 100 closed at **4,997.60** on **3 February 2023**, the last trading day before the earthquakes. During the following week, the index fell sharply as investors reacted to the scale of the disaster, triggering two market-wide circuit breakers. By the time trading was suspended, the market had fallen by roughly **16%**. When trading resumed on **15 February**, the index reopened approximately **7%** below its pre-disaster level before reaching a reported low of **4,505.30**. The BIST 100 returned to its pre-disaster level on **16 February**, making it

the fastest recovery among the three case studies.



The government's emergency response faced significant challenges during the first few days after the disaster. Severe winter weather, damaged transport networks, and disrupted communications slowed search-and-rescue operations across the affected provinces. As emergency infrastructure improved, field hospitals, temporary shelters, and large-scale relief operations were established, while reconstruction efforts began shortly afterwards.

Table 3

Türkiye Emergency Response Timeline

Time After Earthquake	Government and Institutional Action
First 5 hours	Initial damage assessments delayed by severe weather and communication failures
First 24 hours	Major transport routes blocked; emergency response overwhelmed across affected provinces

4 days	Field hospitals, temporary shelters, and heavy equipment deployed across the disaster zone
2 weeks	Large-scale search-and-rescue operations conclude; reconstruction planning begins

A key difference between Türkiye and the other two case studies was the response of financial regulators. On **8 February**, Borsa Istanbul suspended trading in equities, futures, and options for the first time in twenty-four years. All trades executed that day were cancelled, and regulators introduced a temporary ban on short selling before trading resumed on **15 February**. Unlike Japan and the United States, where market prices adjusted continuously after the disasters, Türkiye's market decline was interrupted by direct regulatory intervention.

Monetary policy also became more accommodative following the earthquakes. On **23 February 2023**, the Central Bank of the Republic of Türkiye reduced its benchmark interest rate from **9% to 8.5%**, signalling support for economic recovery. The Turkish government later committed approximately **\$103.6 billion** to reconstruction and recovery, equivalent to about **9% of the country's GDP**, representing the largest fiscal commitment relative to economic size among the three countries examined.

Türkiye entered the disaster with economic conditions that differed substantially from those of Japan and the United States. Although government debt remained relatively low at around **31.5% of GDP**, the country was already experiencing inflation above **60%** and significant currency depreciation. These existing economic pressures may have influenced investor expectations before the earthquakes occurred.

Sectoral impacts varied considerably across the economy. Construction and basic materials companies benefited from expectations of large-scale rebuilding, while transport, manufacturing, and regional businesses suffered immediate losses. Pandey et al. (2024) found that although listed firms experienced significant negative abnormal returns immediately after the disaster, many sectors recovered quickly, with Basic Materials companies recording positive abnormal returns as investors anticipated increased demand during reconstruction.

Türkiye's catastrophe insurance system also limited the role of private insurers in covering disaster losses. The Turkish Catastrophe Insurance Pool (TCIP/DASK) provided mandatory earthquake insurance, but compensation covered only a small proportion of total losses. According to the World Bank (2023), direct physical damage was estimated at **\$34.2 billion**, while insurance covered **less than 5%** of these losses, leaving households and the government responsible for the majority of reconstruction costs.

Unlike the other two case studies, however, Türkiye's market recovery cannot be interpreted solely as evidence of economic resilience. The temporary suspension of trading prevented prices from adjusting freely during the period of greatest uncertainty. As a result, the

reported recovery reflects not only improving investor confidence but also the effects of regulatory intervention. This makes Türkiye a unique case, demonstrating that government action can influence market outcomes not only through fiscal and monetary policy but also by directly shaping how financial markets operate during periods of extreme stress.

Comparative Analysis

Table 4
Market Reaction Summary Across the Three Cases

	Japan (2011)	United States (2005)	Türkiye (2023)
Index	Nikkei 225	S&P 500	BIST 100
Peak-to-trough decline	17.5%	3.7%	9.85%
Days to full recovery	~52	~14	~10*
Trading halted?	No	No	Yes (5 business days)

***Recovery time reflects a five-day exchange closure.**

Table 5
Institutional Response Comparison

	Japan (2011)	United States (2005)	Türkiye (2023)
Central bank response	Aggressive monetary easing	Continued rate hikes	Moderate monetary easing
Fiscal commitment	¥25 trillion (~US\$230B)	~US\$120B	~US\$103.6B (~9% of GDP)
Insurance coverage of losses	<20%	Partial (NFIP deficit)	<5%
Debt-to-GDP before disaster	~230%	~57%	~31.5%

Figure 4. Peak-to-Trough Index Decline by Disaster

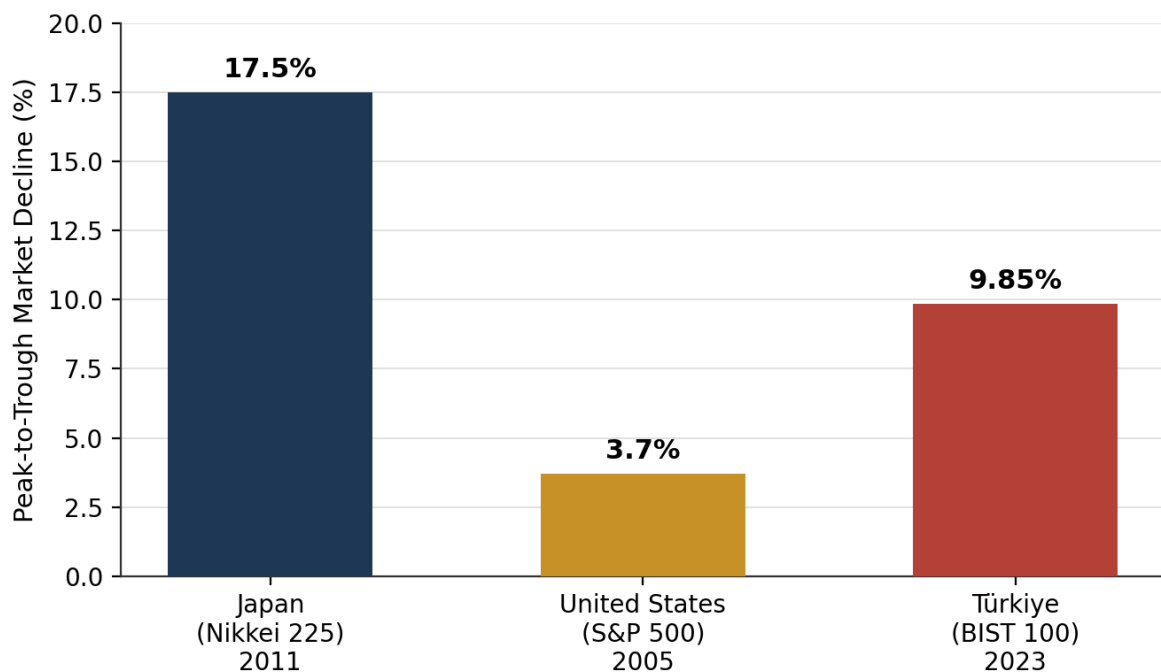
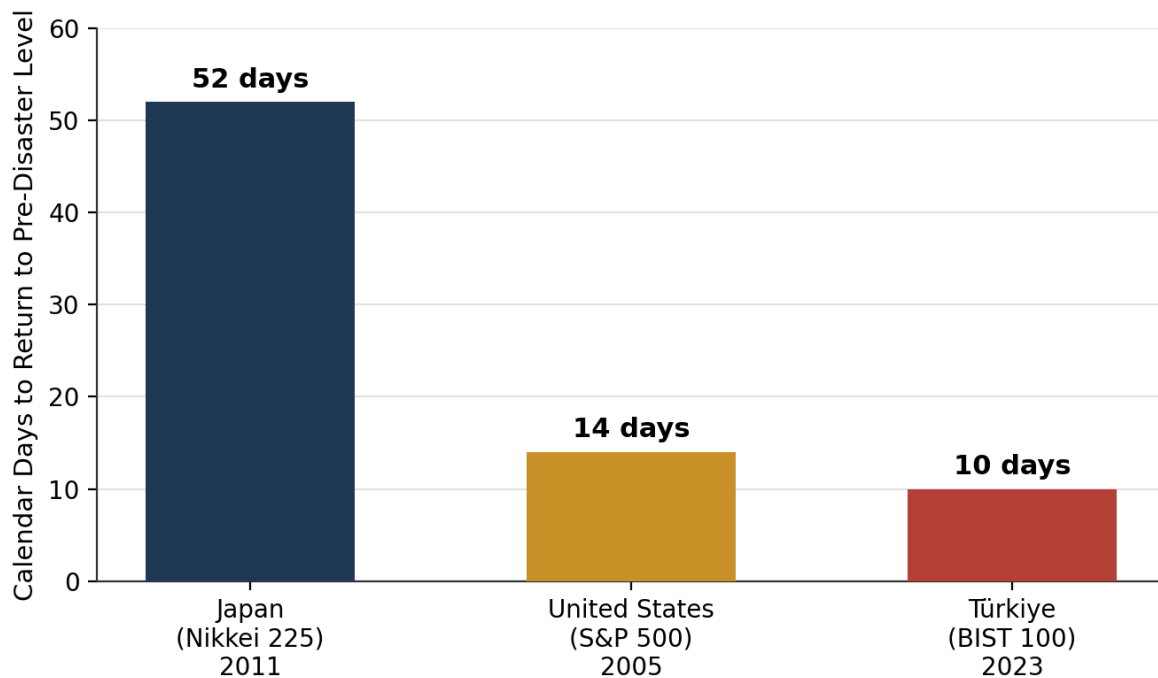


Figure 5. Market Recovery Duration by Disaster



A comparison of the three disasters reveals several clear patterns.

First, **the human impact of a disaster does not appear to predict the severity of the stock market reaction.** Türkiye experienced the highest death toll and the largest number of people affected, yet its stock market decline was smaller than Japan's and its recovery was considerably faster. By contrast, Japan suffered a much lower human toll but experienced the deepest market decline and the longest recovery period. These findings suggest that investors respond primarily to the expected economic consequences of a disaster rather than its humanitarian impact.

Second, **economic damage relative to the size of the economy provides a better explanation for market behaviour than absolute losses.** Although Hurricane Katrina caused approximately **US\$125 billion** in damage, this represented only about **1% of U.S. GDP**, and the S&P 500 recorded the smallest decline of the three markets. Japan and Türkiye both experienced damage equivalent to roughly **3-4% of GDP**, and both recorded substantially larger market declines. This indicates that the economic significance of a disaster depends not only on the amount of damage but also on the economy's ability to absorb it.

Third, **economic structure and sectoral exposure played a major role in determining market resilience.** Japan's disaster directly affected globally important manufacturing industries and the country's nuclear energy sector, disrupting international supply chains and increasing uncertainty for investors. In contrast, Hurricane Katrina was largely confined to the Gulf Coast, allowing the wider U.S. economy to continue operating with relatively limited disruption. Türkiye's most severely affected regions were economically important but did not

include Istanbul, the country's principal financial centre, reducing the impact on many nationally significant firms.

Institutional responses also differed considerably across the three cases, but no single policy measure consistently produced faster market recoveries. Japan responded with aggressive monetary easing and substantial fiscal spending, yet experienced the slowest recovery. The United States continued raising interest rates while recovering relatively quickly. Türkiye combined moderate monetary easing with extensive fiscal support, but its market recovery was also influenced by the temporary suspension of trading. These differences suggest that the effectiveness of policy responses depends on the broader economic context rather than on any individual measure.

Türkiye introduces an additional factor that is absent from the other two case studies: **direct regulatory intervention in financial markets**. The five-day suspension of trading prevented prices from adjusting during the period of greatest uncertainty, meaning that the market's reported recovery cannot be interpreted entirely as a reflection of investor confidence. Unlike fiscal or monetary policy, which aim to influence economic conditions indirectly, the trading halt altered the process of price discovery itself. This makes Türkiye a distinctive example of how regulators can shape short-term market outcomes during periods of extreme volatility.

Finally, the comparison highlights a common feature of all three countries' insurance systems. In each case, catastrophe insurance was designed primarily to maintain the financial stability of insurers rather than fully compensate victims. As a result, governments and households ultimately bore most reconstruction costs, while the insurance sector experienced a disproportionate share of the immediate financial impact—a pattern that is consistent with the findings of Wang and Kutan (2013), Howerton and Bacon (2017), and Feria-Domínguez et al. (2017).

Taken together, these findings provide strong support for **H1**. The evidence shows that the depth and duration of stock market declines are more closely associated with **economic diversification, sectoral exposure, and institutional responses** than with the human or physical severity of the disaster itself. While no three-case comparison can conclusively reject **H0**, the patterns observed across Japan, the United States, and Türkiye are considerably more consistent with the explanation proposed in **H1**.

Discussion

The findings of this study support the central argument that **stock markets respond primarily to economic exposure rather than the human severity of a natural disaster**. Across all three case studies, the size and duration of market declines were more closely associated with economic structure, sectoral concentration, and institutional responses than with the number of deaths or people affected.

The comparison highlights three factors that appear to explain most of the variation in market resilience.

The first is **sectoral exposure**. Japan experienced the deepest and longest market decline because the disaster directly disrupted globally significant manufacturing industries, supply chains, and electricity generation through the Fukushima nuclear accident. By contrast, Hurricane Katrina's economic damage remained largely confined to the Gulf Coast, allowing most of the U.S. economy to continue operating normally. Although Türkiye experienced the greatest human losses, the country's primary financial centre, Istanbul, remained largely unaffected, limiting disruption to many nationally important firms.

The second factor is **economic diversification**. The United States possessed the largest and most diversified economy among the three countries, enabling it to absorb regional damage with relatively limited effects on the broader market. Japan's manufacturing-oriented economy was more exposed to concentrated industrial disruption, while Türkiye entered the disaster with higher inflation and greater macroeconomic uncertainty, making investor confidence more fragile despite its lower government debt.

The third factor is **institutional credibility**. Governments and central banks in all three countries acted to stabilise their economies, but the effectiveness of those responses depended on whether they addressed the specific risks created by each disaster. Japan's large liquidity injections helped maintain financial stability but could not immediately restore disrupted industrial production. The United States recovered quickly despite continuing to raise interest rates because the overall economy remained fundamentally strong. Türkiye's authorities combined fiscal support with direct intervention in financial markets through a temporary trading suspension, demonstrating that regulators can influence market outcomes not only through economic policy but also by temporarily altering the process of price discovery itself.

These findings are broadly consistent with the existing literature. Previous studies found that natural disasters often have limited effects on broad market indices while producing much larger impacts on specific industries and firms (Luo, 2012; Wang & Kutan, 2013; Bourdeau-Brien & Kryzanowski, 2017). The evidence presented here extends those conclusions by showing that cross-country differences in economic structure and institutional capacity also help explain why apparently similar disasters generate very different market outcomes.

Overall, the evidence suggests that investors focus less on the scale of the disaster itself than on its expected economic consequences. Markets appear to assess how severely productive industries will be disrupted, how effectively governments can support recovery, and how quickly normal economic activity is likely to resume. In this sense, stock prices function primarily as indicators of future economic expectations rather than direct reflections of human tragedy.

Limitations

This study has several limitations that should be considered when interpreting its findings.

First, the analysis is based on **three case studies**, which limits the extent to which broad conclusions can be drawn. While the cases were deliberately selected to represent economies with different levels of development and economic structure, they do not provide a sufficiently large sample to establish statistically significant relationships. Consequently, the findings should be interpreted as evidence of recurring patterns rather than definitive proof of causation.

Second, the disasters examined differ in both **hazard type and economic context**. Japan's event involved an earthquake, tsunami, and nuclear accident; Hurricane Katrina was a hurricane and flood; and the Türkiye earthquakes occurred during a period of elevated inflation and macroeconomic instability. These differences make it difficult to isolate the independent effect of any single variable, such as institutional response or sectoral exposure.

A further limitation concerns the measurement of **market recovery**. Returning to a pre-disaster index level does not necessarily indicate that the economy itself has fully recovered. Stock markets reflect investor expectations and broader macroeconomic conditions, both of which may be influenced by factors unrelated to the disaster. Global economic trends, monetary policy, and overall market sentiment may therefore contribute to recovery independently of reconstruction efforts.

Türkiye presents an additional methodological challenge. The five-day suspension of trading by Borsa Istanbul interrupted the normal process of price discovery, meaning that the reported recovery period cannot be interpreted entirely as the result of market forces. While this intervention forms an important part of the analysis, it also limits direct comparisons with Japan and the United States, where trading continued throughout the crisis.

Finally, some disaster statistics-particularly estimates of **economic losses and international aid**-vary across official sources because different organisations apply different methodologies and reporting periods. Where multiple credible estimates existed, this paper relied on widely cited figures and clearly identified their sources. As a result, the numerical values presented should be regarded as reasonable approximations rather than exact measurements.

Future research could strengthen these findings by examining a larger sample of disasters across multiple countries using a consistent event-study methodology. Incorporating additional variables, such as political stability, financial market development, and investor sentiment, would also provide a more comprehensive understanding of the factors influencing post-disaster market behaviour.

Conclusion

This study examined how stock markets responded to three major natural disasters: the **2011 Great East Japan Earthquake**, **Hurricane Katrina (2005)**, and the **2023 Türkiye earthquakes**. By comparing market performance with disaster severity, economic structure, sectoral exposure, and institutional responses, the research sought to determine whether the scale of a disaster alone explains the behaviour of financial markets.

The evidence indicates that **the severity of a disaster in human terms is not the primary determinant of stock market performance**. Although Türkiye experienced the highest death toll and the greatest number of people affected, its market decline was smaller and its reported recovery faster than Japan's. Conversely, Japan experienced the deepest and longest market decline despite suffering significantly fewer casualties. Hurricane Katrina, while causing approximately **US\$125 billion** in damage, had only a limited impact on the broader U.S. stock market because the losses represented a relatively small proportion of the country's highly diversified economy.

Instead, the findings suggest that **economic structure, sectoral exposure, and institutional capacity are more influential in shaping market outcomes than the absolute size of the disaster itself**. Japan's concentration of globally important manufacturing industries and the additional uncertainty created by the Fukushima nuclear crisis contributed to the largest decline. The United States benefited from the size and diversification of its economy, allowing regional disruption to be absorbed with relatively limited effects on the national market. Türkiye demonstrated that regulatory intervention, including the temporary suspension of stock market trading, can significantly influence measured market outcomes alongside fiscal and monetary policy.

Overall, the evidence provides **strong support for the study's hypothesis (H1)**. Across the three case studies, the depth and duration of market declines aligned more closely with **economic diversification, institutional responses, and sectoral concentration** than with casualties or physical destruction alone. While the limited number of case studies prevents definitive causal conclusions, the consistency of these patterns suggests that investors primarily evaluate the expected economic consequences of disasters rather than their humanitarian impact.

The study also contributes to the existing literature by bringing together three disasters from countries with markedly different economic structures within a single comparative framework. Rather than examining only one event or one national market, it demonstrates how differences in economic resilience and policy responses can produce substantially different financial outcomes, even when disasters are similar in magnitude or severity.

Natural disasters will continue to pose significant economic and humanitarian challenges as climate change, urbanisation, and global economic interdependence increase the potential impact of extreme events. Understanding how financial markets respond to these shocks is therefore important not only for investors but also for policymakers seeking to maintain financial stability during periods of crisis. Although future research using larger datasets and formal statistical analysis is needed to test these relationships more rigorously, the evidence

presented here suggests a consistent conclusion: **financial markets price economic resilience more than human tragedy.**